

Technical Report: Implementation of Bio-Sand Filtration and Water Distribution Infrastructure

Location: Northern Shadaw town, Karenni State

Implementing Agency: CTER – Karenni

Reporting Period: February 2 – March 13, 2026

Executive Summary

This report details the implementation of a critical water infrastructure project in the Susee Ywar area of Northern Shadaw. Following significant regional displacement and systemic infrastructure failure due to ongoing conflict, the project successfully restored water access to 79 households and two educational facilities. The initiative integrated community-led construction with the installation of a Bio-Sand filtration system to address both water quantity and quality issues.

Socio-Political Climate

Since the 2021 military coup and the subsequent 2023 "Shadaw Mission" the region has suffered from extreme instability. While original inhabitants have begun returning to peripheral areas, the infrastructure remains largely destroyed or non-functional.

Existing Water Infrastructure Challenges

Prior to the intervention, the village relied on a dilapidated system with the following systemic failures:

- **Infrastructure Deficit:** Only two of six existing storage units were operational.
- **Mineral Obstruction:** Pipelines were heavily clogged by limestone accumulation, a condition worsened by the cessation of maintenance since 2021.
- **External Damage:** Structural integrity was compromised by forest fires and animal interference.
- **Distribution Inefficiency:** Severe leakages along a 5,000-foot transmission line prevented storage tanks from reaching capacity.

Project Methodology and Activities

- Conducted a community meeting to discuss the work plan and promptly formed a committee to ensure the effective implementation and completion of the planned activities.
- Connected water pipe from water sources to village which community participation
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Community Mobilization

Implementation began on February 2, 2026, with the formation of a local committee to oversee the work plan. The project emphasized local ownership, with community members contributing timber, sand, and stones.

Filtration and Storage Installation

The project utilized a multi-stage Bio-Sand filtration approach:

- **Filtration System:** Three 1,000-liter and one 500-liter tank were installed.
- **Layering Technique:** The primary tank utilized a graded media filter consisting of large/medium/small stones, charcoal, and rough/fine sand. Subsequent tanks utilized similar layers without charcoal.
- **Foundation:** A cement-based platform was constructed to support the filtration units.

Pipeline Installation

In March 2026, a new pipeline was connected from the mountain source to the village. To protect the system, a small storage structure was built at the source to regulate flow and mitigate debris entry. Maintenance measures included digging trenches and clearing undergrowth to prevent fire damage.

Beneficiary Demographics

The project directly serves a displaced and returning population:

- **Total Population:** 442 individuals (189 male; 253 female).
- **Educational Support:** 248 students (131 male; 117 female) across a middle school and a primary school.

Infrastructure Improvements

- **Enhanced Storage:** Three existing tanks were restored, and three new units were installed.
- **Accessibility:** Two new distribution points were established, reducing collection time for households.

Challenges and Sustainability

Despite the project's success, several external constraints were identified:

- **Supply Chain Constraints:** Due to the conflict, materials had to be sourced from SAC-controlled areas.
- **Material Quality:** The high water pressure from the mountain source led to the breakage of lower-quality pipes currently available.
- **Capacity Building:** To ensure sustainability, five community members received technical training in pipe connection and repair to facilitate independent maintenance.

Recommendations

To ensure the long-term viability of the water system, it is recommended that the current low-quality piping be replaced with high-quality materials procured from Thailand to withstand the high-pressure environment of the mountainous terrain.

Map of Water Sources

